

UPLIGHT: A Novel Portable Game Device with Omnidirectional Projection Display

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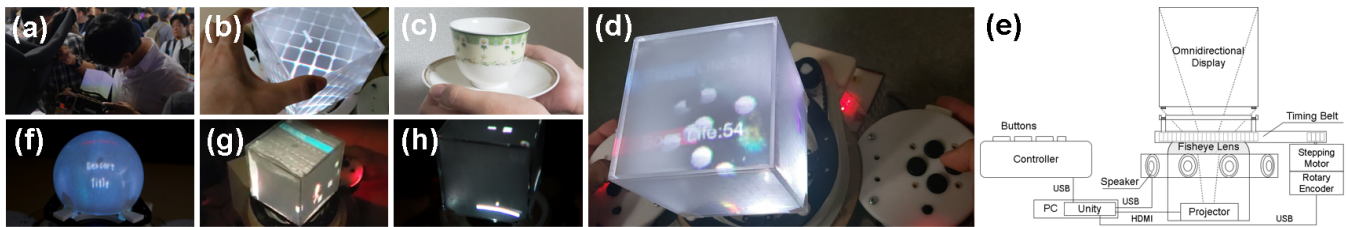


Figure 1: (a) User looking at a hidden part. (b) User holding and rotating a display. (c) Teacup on a saucer. (d) Button controller. (e) Hardware configuration. (f) Sphere-shaped display. (g) Application: action game. (h) Application: block-breaking game.

ABSTRACT

We hypothesized that the act of actively moving one's body to see the hidden parts of a sphere, cube, or any other structure with height and sides would be entertaining. In this paper, we propose a novel portable game device with omnidirectional display called "UPLIGHT," which was created by combining the element of entertainment with the play style of a portable game device. We also describe the design of a prototype and a playable game application that we developed to achieve this interaction.

CCS CONCEPTS

• Human-centered computing → Displays and imagers.

KEYWORDS

entertainment, game, omnidirectional display, tangible interaction, display, controller

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1 INTRODUCTION

An omni-directional display shows images on the entire surface of a three-dimensional (3D) structure. It is capable of visual expression using its shape and tactile expression using physical entities [2] [7]. In addition, because an omnidirectional display does not require special equipment such as an HMD or AR glasses, it is easy for multiple people to share the display [6].

Lin et al. have developed a multi-user social network using a cylindrical omni-directional display, based on the fact that people tend to gather around a circle. They also proposed a multi-user social game using a cylindrical omnidirectional display [5], and Lakmal et al. proposed a game experience using a cylindrical omni-directional display that looks like a magic wand [8].

In an omni-directional display, the user cannot see the entire image from a single viewpoint. In other words, an omni-directional display has a front side and side opposite to the user's viewpoint, which is an invisible or dead space. This dead space has been treated as a limitation of omni-directional displays. For example, proposals have been made to eliminate blind spots by sharing a display with multiple users or using transparent screens [1] [3] [4].

However, there has not been enough discussion focusing on single-user interaction with an omni-directional display with hidden areas. It is believed that hidden areas stimulate users' curiosity and induce them to actively seek to reveal the hidden areas. We hypothesized that the active behavior of users to view the hidden parts has high entertainment value. On the basis of this hypothesis, we propose a hand-held single-user omni-directional display device called "UPLIGHT." In this paper, we describe our thoughts on the design of the proposed device, including suitable shapes, input methods, and necessary functions.

2 SYSTEM CONCEPT

First, we focused on a viewing style in which the user holds with both hands an omnidirectional display. Images are displayed on the entire surface of a 3D structure that can be placed on a desk or held in both hands in almost the same orientation because of its ease of placement and holding. This style of viewing is similar to that of holding a hand-held human figure with both hands and viewing it from various directions. When viewing such a 3D structure, the user has three main ways to view the hidden parts. The first is for the user to turn a light object over to immediately see its back side without changing his or her position, the second is for the user to lean forward to look at the back side of a heavy object (Fig. 1(a)), and the third is for the user to hold the object with one hand and rotate it to reveal the other side carefully without moving the top and bottom (Fig. 1(b)). These viewing methods involve many bodily movements, and we believe that they can enhance the active nature of the information-seeking act.

To create a top and bottom in the omnidirectional display, we focused on the relationship between a flat surface and a 3D structure such as a teacup (Fig. 1(c)). In other words, we thought that by adding a horizontal plane to the bottom of the omnidirectional display, the flat part would be regarded as the bottom. In addition, we noticed that the style of browsing in which each end of the flat part is held with one hand with the display part in the middle is similar to the operation of portable game devices, which have long been popular. We decided to incorporate a controller with a set of buttons divided into left and right groups, similar to handheld game devices (Fig. 1(d)). Our aim was for the proposed device to be recognized as an extension of portable game consoles, which can be used by anyone without any special instructions.

There are many types of entertainment, but we believe that the entertainment we propose in our interaction method comes from the diversity of emotions such as surprise, discovery, and excitement caused by the active search for information. In addition, We believe that moving one's body can also bring about enjoyment.

3 RELATED WORK

In the past, there have been entertainment applications that take advantage of the feature that a omnidirectional display has parts hidden from the user's view. Julie et al. were the first to propose a multi-user competitive and cooperative game using the private side and public top of a spherical omnidirectional display [10]. Subsequently, Ryouta proposed a wireless controller-based competitive game that takes advantage of the feature that an opponent can see a part of the cylindrical omnidirectional display that the player cannot see [9]. However, there has been no work that focuses on single-user interaction.

In addition, Polytron Corp.'s Fez, which can be played on a flat screen, has a horizontal rotation function of the viewpoint. However, this rotation, which is an important part of the information-seeking process, is performed by buttons that can be operated with the fingertips. By contrast, we believe that it is necessary to incorporate actions that require the movement of the entire body in the information-seeking process to enhance surprise and excitement.

4 IMPLEMENTATION

The hardware configuration of the prototype is shown in Fig. 1(e).

The proposed device consists of an omnidirectional projection unit, a display rotation mechanism, a button controller, and a group of speakers arranged in a circle. The projection unit consists of a small projector (Smart Beam Laser), wide-angle lens, and 10 cm square replaceable display. The rotation mechanism consists of a stepping motor with a rotary encoder (PKP244MU12A), timing belt, pulley, and microcontroller for control.

In the projection unit, the image from the small projector passes through a wide-angle lens directly above it and is rear-projected from inside the display. The image of the Unity application is calibrated using the internal and external parameters of the projector and lens distortion parameters. For the display part, we prepared a cube and a sphere 10 cm in diameter (Fig. 1(f)). The former was made by attaching tracing paper to the inside of a 2-mm thick acrylic cube. The latter was made from the outer enclosure of a 10-cm diameter LED light bulb.

5 APPLICATION

To achieve the proposed interaction, game applications need to be designed considering the following elements. First, in terms of visual expression, we consider the continuous connection of game images on the entire display and the movement of objects along the shape of the display. In terms of design, we should employ games designs that create surprises and discoveries when users act to see hidden objects in 3D shapes, game designs that make users want to see many different surfaces, UI designs such as a status display on the top surface, and appropriate level designs based on the assumption that the game will be more difficult because some objects are hidden. We created an action game and a block-breaking game for UPLIGHT that take the above factors into consideration.

The first game is a side-view action game redeveloped for UPLIGHT (Fig. 1(g)). In this game, a character can walk around all sides of the display, including the part hidden from the user. The character moves to the hidden part and changes his or her posture to look at it. In addition, because the character moves around the entire perimeter quickly, the above experience can be quickly repeated. In this application, the top surface is used as a UI to display the parameters of the character and menus, and when an enemy fires a bullet at the top surface, the bullet passes through the top surface and hits the character.

The second is a traditional block breaking game, modified for UPLIGHT (Fig. 1(h)). In this game, the ball flies along the shape of the block, giving a unique trajectory to the block-breaking game. Each time we change the shape of the 3D object, the ball takes a different trajectory, changing the enjoyment and experience. It is difficult to completely predict the trajectory of a ball that flies into an invisible area, and thus the user moves his or her body to look at the invisible area and is surprised by the unexpected position and trajectory of the ball, which is an interesting experience brought about by the occlusion in the omnidirectional display.

6 FUTURE WORK

In future, we would like to verify the entertainment value of the act of revealing the hidden part of the omnidirectional display

by improving the prototype and implementing the application. In addition, we would like to use sounds and smells to enhance the active nature of the user's search for information.

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